

O-band InP Mach-Zehnder Modulator for 400 Gb/s per Lane in a Generic Pure-Play Foundry

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Abstract To address increasing datacenter requirements of 400 Gb/s per lane we present an InP Mach-Zehnder modulator with measured electro-optic bandwidth above 67 GHz and V_π of 2.6–3.4 V for an interaction length of just 0.8 mm. Our physics-based model predicts operation beyond 110 GHz. ©2026 The Author(s)

Introduction

The enormous growth in global data traffic - fuelled by AI, cloud computing, and HD video streaming - has forced datacenters to scale drastically over the last several years. To meet future bandwidth demands, the industry is transitioning from 400G toward 1.6T architectures. This drives the necessity for high-speed optical transmitters at rates that cannot be supported by Silicon anymore. InP is the material of choice to address this requirement.

Current datacenter requirements demand modulators that are not only high-speed but also maintaining stable performance across wide operating temperature ranges. Among competing technologies, the Mach-Zehnder modulator (MZM) on Indium Phosphide (InP) stands out for its efficiency, small footprint, high extinction ratios and potential for monolithic integration with active components [1-4]. Furthermore, the MZM offers operation across CWDM wavelength range needed for integration in FR4 architecture.

In this paper, we present an integratable InP-based MZM designed for next-generation interconnects. Our small-signal measurements demonstrate operation well beyond 67 GHz, showing strong agreement with our simulation model. Furthermore, the simulation model predicts bandwidths exceeding 100 GHz, confirming the platform's suitability for 400 Gb/s per lane transmission using PAM4 signalling

Design

Our InP modulator is implemented as a Mach-Zehnder interferometer in which each arm consists of a p-i-n heterostructure defined in a deep-etched ridge waveguide. The p-type layer is electrically contacted to travelling-wave electrodes (TWEs) located on top of the heterostructure. The two signal electrodes are placed in close proximity, forming a coupled coplanar-stripline transmission line, while two ground electrodes are added on the outer sides. As shown in Fig. 1, the device is optimized for differential drive using a GSSG pad configuration, and the transmission line is

terminated on chip through a layout-tuneable resistor.

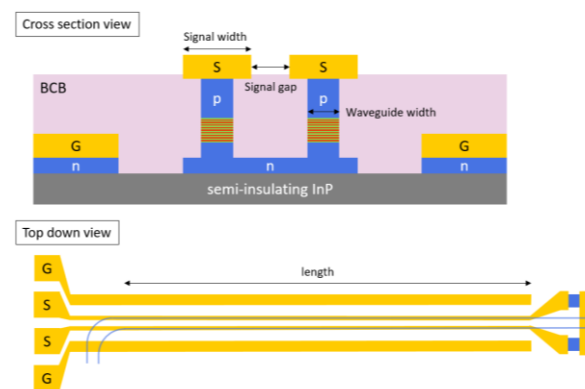


Fig. 1: Schematic cross-section view of the p-i-n InP TWE-MZM (top) and its top-down view (bottom)

Within the SMART Photonics PDK, four MZM parameters can be adjusted by the designer: signal width, signal gap, waveguide width, and interaction length, as illustrated in Fig. 1. The signal width and signal gap primarily determine the transmission-line impedance and electro-optic (EO) bandwidth. The waveguide width also affects these quantities, but at the expense of increased optical insertion loss. Finally, the interaction length sets the trade-off between the half wave voltage V_π and EO bandwidth.

To optimize the device performance based on the above-mentioned layout parameters and the heterostructure doping profile, we developed a physics-based model with zero fitting parameters using the following workflow (see Fig. 2):

1. Solve Poisson-drift-diffusion equations in 1D for our p-i-n heterostructure under 5V reverse bias. Obtain the depleted region width and the new free carrier densities at the cladding layers.
2. Build the substrate definition in the 3D FEM software, modelling the depleted region as a dielectric. The semiconductor layers are assigned a conductivity and dielectric constant based on their carrier

- density, following [5].
3. Create the layout and map the substrate to generate a full 3D model. Run 3D FEM simulation to obtain S-parameters, for several layout parameter combinations.
 4. Based on S-parameters, we compute the microwave transmission line parameters to feed the EO response model.
 5. Finally, we compute the EO response as a function of frequency based on Ghione's formalism [6].

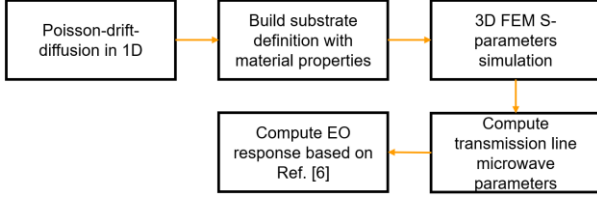


Fig. 2: Flow diagram for the simulation of electro-optic (EO) S-parameters.

Results

Owing to the trade-off between V_π and EO bandwidth due to the interaction length, we first report an optimized multi quantum well (MQW) epitaxy design achieving half-wave voltages as low as $< 2 \text{ V} \cdot \text{mm}$. The main phase-shifting mechanism of our InAlGaAs/InAlGaAs MQW epitaxy stack is the quantum-confined Stark effect (QCSE) [7].

In Fig. 3 we demonstrate a low $V_\pi \cdot \text{mm}$ for a wide range of wavelengths and temperatures, as well as a static extinction ratio (ER) above 20 dB. We report our results as a function of detuning, i.e. the difference between the laser wavelength and the photoluminescence (PL) peak wavelength of the MQW. This helps the user select their operation wavelength range by choosing the PL peak wavelength of the MQW stack. The MZM response showed on the left panel in Fig. 3 was measured at 1310 nm wavelength, corresponding to a detuning of 98 nm. As expected for the InAlGaAs/InAlGaAs system, we observe a strong

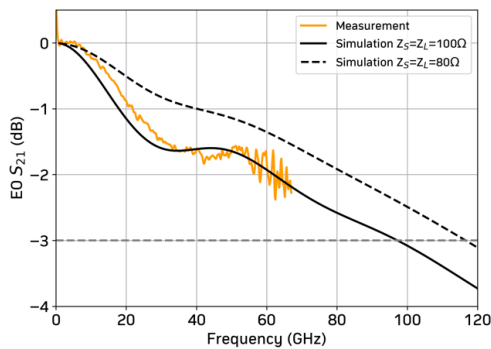


Fig. 4: Differential electro-optic S-parameters for a device of 0.8 mm interaction length, at 5 V reverse bias. S-parameters were obtained at 20 °C under a laser wavelength of 1310 nm. The corresponding half-wave voltage V_π for this interaction length varies from 2.6 to 3.4 V, depending on the temperature.

$V_\pi \cdot \text{mm}$ dependence on wavelength and temperature. A PL peak wavelength red shift as function of temperature of $0.5 \text{ nm}/^\circ\text{C}$ was measured.

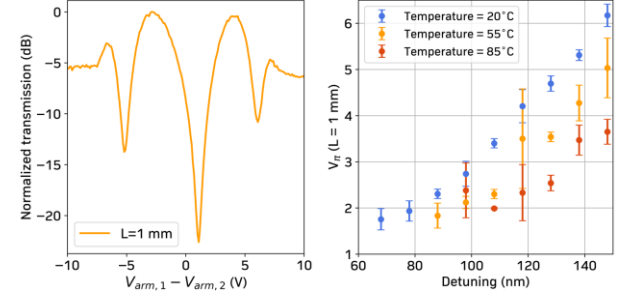
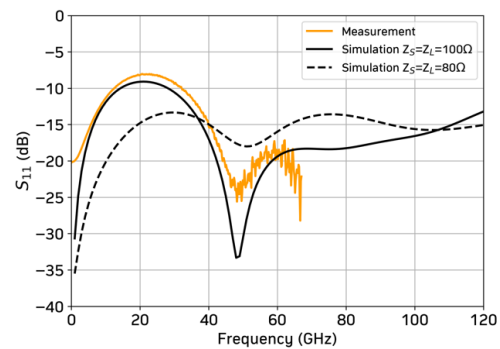


Fig. 3: Mach-Zehnder interferometer DC response at 20 °C and 1310 nm wavelength (left panel) and V_π at various temperatures as a function of detuning, where 98 nm detuning corresponds to 1310 nm wavelength for the devices in this work (right panel). In both panels the results are for a 1 mm long modulator. V_π is measured with both arms reversed bias at a voltage such that the extra voltage-dependent optical insertion loss remains below 2 dB, and at all cases at a maximum reverse bias of 5 V.

We estimate the optical insertion loss of the MZM under zero reverse to be $< 6 \text{ dB}$.

The low V_π achieved enables the use of a compact MZM with an interaction length of just 0.8 mm. At 1310 nm, the corresponding device exhibits a V_π between 3.4 V at 20 °C and 2.6 V at 85 °C. Guided by our model, we selected a signal width of $10 \mu\text{m}$, a signal gap of $13 \mu\text{m}$ and a waveguide width of $1.0 \mu\text{m}$. The transmission line is terminated on chip with a 100Ω resistor.

Fig. 4 shows the measured differential EO S-parameters for the selected 0.8 mm device. Because the vector network analyser (VNA) is limited to 67 GHz, the 3-dB EO bandwidth cannot be directly extracted from the measurement. Nevertheless, the measured EO response exhibits only 2 dB attenuation at 67 GHz, indicating that the 3-dB point lies beyond the available measurement range. To support the claim of a higher operation speed than our VNA maximum frequency, we plot on the same graph the EO response predicted from our model, yielding an excellent match to the



experimental data with no fitting parameter. The model was also verified with more device parameter combinations, yielding again an excellent match to the measured data. Based on this model our MZM is expected to reach 97 GHz bandwidth.

The measurements were performed on a 100 Ω differential calibrated setup. Since 100 Ω source/load impedance is not matching the characteristic impedance of our transmission line, we used our model to evaluate operation under 80 Ω source/load impedance. Under these conditions, the predicted bandwidth increases to 116 GHz. These results show the applicability of this device for 400 Gb/s per lane PAM4 transmission.

Conclusions

In this work, we demonstrated an O-band InP MZM suitable for 400Gb/s per lane. Our model, verified with experimental data, shows capability of reaching >110 GHz under 80 Ω impedance, with a V_{π} as low as 2.6-3.4 V at 1310 nm, working from 20 °C up to 85 °C. The low V_{π} combined with a static ER above 20 dB ensures a low peak-to-peak drive voltage (V_{pp}), highlighting the advantage of our InP MZM over competing architectures such as electro-absorption modulators (EAMs).

Another important advantage of our solution over other approaches is its compactness. Our InP MZM has an interaction length of just 0.8 mm, while most solutions based on its thin-film lithium niobate (TLFN) counterpart use multi-mm to cm-scale phase sections [8-9].

Moreover, we built a physical model with zero fitting parameters capable of reproducing our EO high-frequency measurements with an excellent accuracy. This model will help improving further our MZM to reach even higher modulation speeds.

In summary, these results show that our InP MZM, available in a generic pure-play foundry, meets the requirements for post-silicon photonics and 400 Gb/s per lane modulation speeds.

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